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Adjustable locking pliers

Abstract

An adjustable locking pliers includes a stationary handle, a stationary jaw, an operating lever and a movable jaw. The jaw is coupled by an axis which is selectively displaceable to alter the jaws' gripping capacity. The operating lever is coupled to the movable jaw by a first pivot. The lever rear forms a movable handle. A spring biases rotation of the movable jaw about the axis away from the stationary jaw. An actuation rod having a series of teeth and a stop face is coupled to the operating lever by a second pivot part way along the operating lever and coupled to the stationary handle by means of a third pivot. A catch having a series of teeth and a stop face is pivotally coupled to the operating lever. The teeth of the actuation rod and the teeth of the catch are configured to engage each other to permit multiple clamping positions.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This patent application claims priority to EP19185834.9, filed Jul. 11, 2019, which is hereby incorporated by reference in its entirety.

FIELD OF THE INVENTION

(2) The present invention relates to an adjustable locking pliers of the type that allow a strong clamping force to be exerted for a long period of time, without intervention from the operator, on an object, or, more frequently, on two objects which need to be held together. More specifically, the present invention relates to an improved adjustable locking pliers having an improved latch lock mechanism capable of allowing the pliers to achieve multiple clamping positions by simply closing the handles.

BACKGROUND

(3) Locking pliers are well known in the art. Patent publication number EP2826595 discloses a design of adjustable locking pliers that have been in production for several years. These locking pliers have advantages over previous models of locking pliers. For example, they have increased gripping capacity. They also avoid explosive reaction to the hands of the operator from the elastic energy contained within the clamped pliers. Despite these improvements, the locking pliers of EP2826595 experience the drawback of only having a single locking or clamping position. This limitation is due to the design of the lock latch mechanism.

(4) The latch lock mechanism **116** of EP2826595 comprises an operating lever **124**, the front end of which is coupled to the lower vertex of the movable jaw **114** by a first pivot **126** formed by a first axle. The midpoint of the operating lever **124** is coupled to the front end of an actuation rod **128** by a second pivot **130** formed by a second axle. The rear end of the operating lever **124** is elongate and forms a second, movable handle **132** situated underneath the stationary handle **110**. Referring in particular to FIG. 1, the movable handle **132** is manually pivotable toward the stationary handle **110**, in the direction of arrow A, and away from the handle **110**, in the direction of arrow B.

(5) As shown in FIG. 3, the front end of the actuation rod **128** has an engagement tooth **152**, the front face of which forms an upper arc **154** of a circle that is centered on the second pivot **130**, a lower arc **156** of a circle of smaller radius also centered on the second pivot **130**, and a radial face **158** which connects the two arcs **154,156**. The radial face **158**, which constitutes an engagement face, extends substantially radial with respect to the second pivot **130**.

(6) The latch lock mechanism **116** also comprises a locking/unlocking catch **160** coupled to the operating lever **124** by an axle **162** located close to the second pivot **130**. On its inner side, facing upwards towards the stationary handle **110**, the catch **160** has a recess **164** delimited at the top by an upper triangular engagement tooth **166** and at the bottom by a lower triangular stop tooth **168**. The upper face **169** of the stop tooth **168**, which constitutes a stop face, extends substantially radial with respect to the axis of rotation of the axle **162**.

(7) The latch lock mechanism **116** comprises a traction spring **170** hooked under tension between a finger **172** on the operating lever **124** (located just behind the first pivot **126**) and an eyelet **174** in the actuation rod **128** (located just above the second pivot **130**).

(8) The latch lock mechanism **116** comprises a double torsion spring **176** mounted upon the axle **162** and fixed to the catch **160**. A protruding part **176a** of the double torsion spring **176** acts upon the operating lever **124** such that the catch **160** is biased by the double torsion spring **176** to rotate in a clockwise direction E about the axle **162**, as is shown in FIG. 3. The catch **160** forms a trigger which protrudes slightly from the movable handle **132**.

(9) The latch lock mechanism **116** comprises a single torsion spring **178** the rear end **178a** of which is mounted upon a finger **180** on the intermediate connecting part **112**. A middle coil **178b** of the single torsion spring **178** is unattached. A forward protruding part **178c** of the single torsion spring **176** acts upon the dowel **120** such that the dowel **120** is biased by the single torsion spring **178** in a generally forward direction of arrow F, as is shown in FIG. 2. The forward bias of the single torsion

spring **178** is only just enough to reliably retain the dowel **120** in a notch **150a-150e** when the jaws are unclamped.

(10) When the locking pliers **102** is not in use, tension in the traction spring **170** pulls the second pivot **130** away from an imaginary line IL between the first **126** and third **148** pivots. This reduces the distance between the first **126** and third **148** pivots which causes the movable jaw **114** to pivot about the dowel **120** (in the anti-clockwise direction of arrow G) away from the stationary jaw **108** and the movable handle **132** to pivot about the first pivot **126** (in the direction of arrow B) away from the stationary handle **110**. It also causes a reduction in a locking angle α enclosed by the first **126**, second **130** and third **148** pivots. The jaws **108,114** are either in, or moving towards, an open position like that shown in FIG. 4. The radial face **158** of the actuation rod's engagement tooth **152** rests against the stop face **169** of the catch's stop tooth **168** and the catch's engagement tooth **166** rests against the upper arc **154** of the front face of the actuation rod **128**. This prevents the two handles **110,132** from moving further apart from one another.

(11) The lower part of the catch **160** forms a trigger which protrudes slightly beneath the movable handle **132**. When the locking pliers **102** are in use, the operator, using all four fingers, begins to pull the movable handle **132** closer to the stationary handle **110** (in the direction of arrow direction A) which is wedged firmly in the palm of the operator's hand. The second pivot **130** moves towards the imaginary line IL between the first **126** and third **148** pivots. This increases the distance between the first **126** and third **148** pivots which causes the movable jaw **114** to pivot about the dowel **120** (in the clockwise direction of arrow H) towards the stationary jaw **108** and the movable handle **132** to pivot about the first pivot **126** (in the direction of arrow A) towards the stationary handle **110**. The locking angle α gradually widens, and the movable jaw **114** rotates about the dowel **120**. Thus, the jaws **108,114** move towards a closed position like that shown in FIG. 1 and begin to clamp an object. Tension in the traction spring **170** gradually increases. At the same time, the tip of the actuation rod's engagement tooth **152** moves closer to that of the catch's engagement tooth **166** while contact between the tip of the catch's engagement tooth **166** and the upper arc **154** of actuation rod's engagement tooth **152** and between the tip of the catch's stop tooth **168** and the actuation rod's lower arc **156** is sustained by the bias of the torsion spring **176**. Throughout this movement, the catch **160** is immobilized with respect to the movable handle **132** so that the trigger may form a purchase for the operator's index finger.

(12) By continuing to move the two handles **110,132** closer together, the operator firmly clamps the object between the jaws **108,114** and slightly increases the locking angle α to a degree at which the actuation rod's engagement tooth **152** snap-fastens behind the catch's engagement tooth **166**, as is best shown in detail by FIG. 3. The upper face of the catch's engagement tooth **166** is substantially radial with respect to the second pivot **130**. Stresses due to clamping plus tension in the traction spring **170** tend to reduce the locking angle α . However, the tip of the catch's engagement tooth **166** abuts the actuation rod's radial face **158** to prevent a reduction of the locking angle α . The locking pliers **102** has now reached a stable clamped position. The centre of the second pivot **130** has not passed the imaginary line IL between the first **126** and the third **148** pivots. The locking angle α is slightly less than 180 degrees. Typically the locking angle α is in the order of 170 to 175 degrees. In this position, the trigger protrudes further beneath the movable handle **132**. The snap-fastening can be felt by the operator's index finger.

(13) The double torsion spring **176** urges the catch **160** to pivot in a clockwise direction E about the axle **162**. The engagement teeth **152,166** abut each another with a force which is substantially perpendicular to the imaginary line IL between the first **126** and third **148** pivots. If the operator pulls the two handles **110,132** even closer together, the movable handle's abutment point **184** will move into abutment with the actuation rod's abutment point **182** before the second pivot **130** passes the imaginary line IL between the first **126** and the third **148** pivots. This guarantees that the second pivot **130** never passes the imaginary line IL. Once the operator releases the movable handle **132**, the locking pliers **102** adopts the stable clamped position under the bias of the traction spring **170**.

The lock mechanism **116** exerts a locking force LF in a clockwise direction about the dowel **120** which is counteracted by a clamping force CF between the jaws **108,114**.

(14) To release the object clamped between the jaws **108,114**, the operator takes hold of the locking pliers **102**, pulls the handles **110,132** slightly closer together until the movable handle's abutment point **184** moves into abutment with the actuation rod's abutment point **182**. The operator presses a finger on the trigger **160** to disengage the engagement teeth **152,166** and then releases the movable handle **132**

(15) to allow the jaws **108,114** to open automatically. This opening is caused by tension in the traction spring **170** which tends to pull the first **126** and third **148** pivots together, push the second pivot **130** away from the imaginary line IL and rotate the movable jaw **114** away from the stationary jaw **108** in the anti-clockwise direction of arrow G. The most wide-open position is delimited by abutment between the catch's stop tooth **168** and the actuation rod's radial tooth **158**.

(16) Thus, an operator using the locking pliers of EP2826595 has to engage in trial & error in order to find the appropriate clamping setup position. This can be a time consuming process. It would be advantageous to have locking pliers with multiple intermediate locking positions, that would allow an operator to more quickly set up the locking pliers in an appropriate locking or clamping position. The locking pliers of the present invention include all of the advantages of EP2826595, while simultaneously providing the added benefit of allowing for multiple incremental locking or clamping positions.

SUMMARY OF THE INVENTION

(17) In at least one embodiment of the present invention, there is provided a locking pliers including a stationary assembly having an elongated overall shape, wherein a rear end of the stationary assembly forms a stationary handle and a front end of the stationary assembly forms a stationary jaw. The locking pliers further include a movable assembly having an operating lever and a movable jaw, wherein the movable jaw is pivotally coupled to the stationary assembly by a jaw axis to enable clamping of an object between the movable and stationary jaws and wherein the operating lever has a front end that is pivotally coupled to the movable jaw by a first pivot and the operating lever has a rear end that forms a movable handle. The locking pliers also include a spring for biasing rotation of the movable jaw about the jaw axis away from the stationary jaw. Also included is an actuation rod having a series of teeth and a stop face on its front end, which is pivotally coupled to the operating lever by a second pivot part way along the operating lever, said actuation rod further includes a rear end pivotally coupled to the stationary handle by means of a third pivot and wherein the actuation rod and a portion of the operating lever that extends between the first and the second pivots defines a latch lock mechanism. Also included is a catch pivotally coupled to the operating lever and having a series of the teeth, a stop face and a trigger surface that is configured to be engaged by an operator in order to pivot the catch. The actuation rod and the operating lever each have abutment means and wherein the abutment means are mutually aligned to stop the latch lock mechanism from passing beyond a point of alignment of the first, second and third pivots upon movement of the movable handle toward the stationary handle during clamping. The invention is characterized in that the teeth of the actuation rod and the teeth of the trigger are configured to engage each another to permit multiple clamping positions wherein in each of said positions, the movable jaw and fixed jaw are held in a fixed clamped arrangement.

(18) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms, "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the root terms "include" and/or "have", when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of at least one other feature, step, operation, element, component, and/or groups thereof.

(19) As used herein, the terms "comprises," "comprising," "includes," "including," "has," "having"

or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of features is not necessarily limited only to those features but may include other features not expressly listed or inherent to such process, method, article, or apparatus.

(20) For definitional purposes and as used herein, “connected” or “attached” includes physical or electrical, whether direct or indirect, affixed or adjustably mounted. Thus, unless specified, “connected” or “attached” is intended to embrace any operationally functional connection.

(21) As used herein, “substantially,” “generally,” “slightly” and other words of degree are relative modifiers intended to indicate permissible variation from the characteristic so modified. It is not intended to be limited to the absolute value or characteristic which it modifies but rather possessing more of the physical or functional characteristic than its opposite, and approaching or approximating such a physical or functional characteristic.

(22) In the following description, reference is made to the accompanying drawings which are provided for descriptive and illustration purposes as representative of specific exemplary embodiments in which the invention may be practiced. Given the following description of the specification and drawings, the apparatus, methods, and systems should become evident to a person of ordinary skill in the art. Further areas of applicability of the present teachings will become apparent from the description and illustrations provided herein. It is to be understood that other embodiments can be utilized and that structural changes based on presently known structural and/or functional equivalents can be made without departing from the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above-mentioned and other features and advantages of this invention, and the manner of attaining them, will become more apparent and the invention will be better understood by reference to the following descriptions of embodiments of the invention taken in conjunction with the accompanying drawings:

(2) FIG. 1 shows a side elevation view of locking pliers according to the prior art;

(3) FIG. 2 show a side cutaway view of locking pliers according to the prior art in the closed/clamped position.

(4) FIG. 3 shows a detail VI of FIG. 2.

(5) FIG. 4 shows a side cutaway view of locking pliers according to the prior art in the open/unclamped position.

(6) FIG. 5 shows a side elevation of the locking pliers of the present invention.

(7) FIG. 6 shows an exploded view of the locking pliers of the present invention.

(8) FIG. 7 shows a side cutaway view of locking pliers according to the present invention in a first locked/clamped position.

(9) FIG. 8 shows a detail of VII of FIG. 7.

(10) FIG. 9 shows a side cutaway view of the locking pliers according to the present invention in a second locked/clamped position.

(11) FIG. 10 shows a detail of VIII of FIG. 9.

(12) FIG. 11 shows a side cutaway view of the locking pliers of the present invention in the open/unclamped position.

(13) FIG. 12 shows a detail of IX of FIG. 11.

(14) FIG. 13 shows a detail of X of FIG. 5.

(15) FIG. 14 shows detail XI of FIG. 9.

(16) Corresponding reference characters indicate corresponding parts throughout the several views. The exemplifications set out herein illustrate embodiments of the present invention, and such

exemplifications are not to be construed as limiting the scope of the present invention in any manner.

(17) These and other features of the invention will become apparent upon review of the following description of the present embodiments of the invention, taken into conjunction with the figures.

DETAILED DESCRIPTION

(18) A detailed description of apparatuses, methods, and systems, consistent with embodiments of the present disclosure is provided below. While several embodiments are described, it should be understood that the disclosure is not limited to any one embodiment, but instead encompasses numerous alternatives, modifications, and equivalents. In addition, while numerous specific details are set forth in the following description in order to provide a thorough understanding of the embodiments disclosed herein, some embodiments can be practiced without some or all of these details. Moreover, for the purpose of clarity, certain technical material that is known in the related art has not been described in detail in order to avoid unnecessarily obscuring the disclosure. An example implementation of a locking pliers **103** is shown in FIGS. **1-14**. The example tool **103** illustrated in FIGS. **1-14** is a handheld locking plier, simply for ease of discussion and illustration. However, the principles to be described herein may be applied to other types of handheld tools that are operable in different modes such as, for example, use of the tool without the locking function.

(19) As illustrated in FIGS. **5-7**, the locking pliers **103** is flat in overall shape and consists of an upper stationary assembly **104** and a lower movable assembly **106**. The locking pliers **103** of the present invention are an improvement over the prior art. Thus, they share many of the same elements. For ease of reference, similar elements will share the same reference number.

(20) The stationary assembly **104** is elongate with a front end part constituting a stationary jaw **108**, a rear end part constituting a stationary handle **110**, and an intermediate connecting part **112**. The stationary handle **110** and the intermediate connecting **112** parts are formed of sheet metal shaped in a generally U-shaped cross-section. The stationary jaw **108** is formed of metal with an array of serrations **109** on its inside for gripping an object. The stationary jaw **108** is fastened to the intermediate connecting part **112** by a pair of rivets **113a**, **113b**.

(21) The movable assembly **106** comprises a generally V-shaped movable jaw **114** and a latch lock mechanism **117** equipped with an adjusting device **118**. The movable jaw **114** is formed of metal with an array of serrations **119** on its inside for gripping an object. The rear upper vertex of the movable jaw **114** is coupled to the stationary assembly **104** via a dowel **120**. The dowel **120** is contained within a notched track **122** in the intermediate connecting part **112**.

(22) The latch lock mechanism **117** comprises an operating lever **124**, the front end of which is coupled to the lower vertex of the movable jaw **114** by a first pivot **126** formed by a first axle. The midpoint of the operating lever **124** is coupled to the front end of an actuation rod **129** by a second pivot **130** formed by a second axle. The rear end of the operating lever **124** is elongate and forms a second, movable handle **132** situated underneath the stationary handle **110**. Referring in particular to FIG. **5**, the movable handle **132** is manually pivotable toward the stationary handle **110**, in the direction of arrow A, and away from the handle **110**, in the direction of arrow B.

(23) The adjustment device **118** comprises a knurled cylinder **140** fixed around a threaded nut **141** and a screw **142** which passes through the nut **141**. The cylinder **140** and the nut **141** are coaxial with a longitudinal axis **143** of the screw **142**. The cylinder **140** is accommodated in a recess **144** in the handle **110** opening towards the top of the handle **110**. The nut **141** is axially offset in relation to the cylinder **140** so that a front portion **141a** of the nut **141** protrudes from the front end of the cylinder **140**. The front portion **141a** protrudes beyond the recess **144** and inside the stationary handle **110**. The handle **110** acts as a collar about the front portion **141a** which retains the cylinder **140** and the nut **141** within the recess **144** should the nut **141** ever threadingly disengage from the screw **142**. The rear end of the cylinder **140** has an internal diameter restriction **140a** which abuts the rear end of the nut **141**. The restriction **140a** prevents the nut **141** from withdrawing further rearwardly inside the cylinder **140**.

(24) The cylinder **140** is accessible through the recess **144**. The cylinder **140** and the nut **141** can be manually rotated about the longitudinal axis **143** of the screw **142** whilst being held captive in the recess **144** of the stationary handle **110** and being prevented from translational movement relative to the stationary handle **110**. The rear end of the screw **142** passes, able to rotate freely, through the clevis piece **146** which is coupled to the rear end of the actuation rod **129** by a third pivot **148**. Rotation of the knurled cylinder **140** and the nut **141** causes forward or backward translation of the screw **142**, and the latch lock mechanism **117** coupled thereto, to pivot the movable jaw **114** about the first pivot **126** to perform fine adjustment of the jaws' gripping capacity.

(25) The notched track **122** is a pair of identical mutually aligned notched windows **122a,122b** each window passing through an opposite side faces of the U-shaped intermediate connecting part **112**. Each window **122a,122b** has, on a front side facing the jaws **108,114**, a series of five notches **150a-150e**. There may be at least two notches depending on how wide the jaws are designed to open from each other. Each pair of mutually aligned notches **150a-150e** is suitable for supporting and cradling the dowel **120** when jaws are clamped and the latch lock mechanism **117** is in a locked position. When the jaws are unclamped and the latch lock mechanism **117** is in an unlocked position, the dowel **120** is slideable between notches **150a-150d** to perform coarse adjustment of the jaws' gripping capacity.

(26) The front end of the actuation rod **129** includes portion defining an upper arc **202** that is centered on the second pivot **130**. The upper arc **202** includes a series of teeth **204** and a stop face **206**. Stop face **206** is positioned to extend substantially radially from the second pivot **130**. The front end of the actuation rod **129** further includes a portion defining a lower arc **208**. The lower arc **208**, which is positioned adjacent to the stop face **206**, is also centered on the second pivot **130**. However, the lower arc **208** has a smaller radius than the upper arc **202**.

(27) The latch lock mechanism **117** also comprises a locking/unlocking catch **161** coupled to the operating lever **124** by an axle **162** located close to the second pivot **130**. The catch **161** includes an inner side **210**. The inner side **210** is positioned to be facing upward toward the stationary handle **110**. The inner side **210** includes a recess **212** positioned between series of teeth **214**, and a stop face **216**. Teeth **214** are configured to slidably engage teeth **204** in a ratcheting fashion, wherein the teeth **204, 214** allow rotation of the actuation rod in a first direction (clockwise as shown in the figures) but not in a second direction (anti-clockwise as shown in the figures). Stop face **216** is configured to engage stop face **206** and prevent further rotation of the actuation rod in the first direction.

(28) Catch **161** further includes an outer side that defines a trigger surface **218**. Trigger surface **2018** is configured to be engaged the forefinger of an operator to selectively allow the catch **161** to be pivoted around axle **162**.

(29) The latch lock mechanism **117** comprises a traction spring **170** hooked under tension between a finger **172** on the operating lever **124** (located just behind the first pivot **126**) and an eyelet **174** in the actuation rod **128** (located just above the second pivot **130**).

(30) As best seen in FIG. **10**, the latch lock mechanism **117** comprises a double torsion spring **176** mounted upon the axle **162** and fixed to the catch **161**. A protruding part **176a** of the double torsion spring **176** acts upon the operating lever **124** such that the catch **161** is biased by the double torsion spring **176** to rotate in a clockwise direction E about the axle **162**. The trigger surface **218** of the catch **161** protrudes slightly from the movable handle **132**.

(31) The latch lock mechanism **117** further includes a single torsion spring **178** the rear end **178a** of which is mounted upon a finger **180** on the intermediate connecting part **112**. A middle coil **178b** of the single torsion spring **178** is unattached. A forward protruding part **178c** of the single torsion spring **176** acts upon the dowel **120** such that the dowel **120** is biased by the single torsion spring **178** in a generally forward direction of arrow F, as is shown in FIG. **7**. The forward bias of the single torsion spring **178** is only just enough to reliably retain the dowel **120** in a notch **150a-150e** when the jaws are unclamped.

(32) When the locking pliers **103** is not in use, tension in the traction spring **170** pulls the second pivot **130** away from an imaginary line IL between the first **126** and third **148** pivots. This reduces the distance between the first **126** and third **148** pivots which causes the movable jaw **114** to pivot about the dowel **120** (in the anti-clockwise direction of arrow G) away from the stationary jaw **108** and the movable handle **132** to pivot about the first pivot **126** (in the direction of arrow B) away from the stationary handle **110**. It also causes a reduction in a locking angle α enclosed by the first **126**, second **130** and third **148** pivots. The jaws **108,114** are either in, or moving towards, an open position like that shown in FIG. **11**. In this open position, as best seen in FIG. **12**, the stop face **206** of the actuation rod **129**, is in direct engagement against the stop face **216** of the catch **161**. To accommodate this position, the teeth **214** of the catch are completely disengaged from the teeth **204** of the actuation rod **129**. Recess **212** provides the clearance to allow teeth **204** to avoid contact with the catch **161**, while simultaneously permitting the stop faces **206, 216** to be engaged to one another. This prevents the two handles **110, 132** from moving further apart.

(33) The lower part of the catch **161** forms a trigger surface **218** that protrudes slightly beneath the movable handle **132**. When the locking pliers **103** are in use, the operator, using all four fingers, begins to pull the movable handle **132** closer to the stationary handle **110** (in the direction of arrow direction A) which is wedged firmly in the palm of the operator's hand. The second pivot **130** moves towards the imaginary line IL between the first **126** and third **148** pivots. This increases the distance between the first **126** and third **148** pivots which causes the movable jaw **114** to pivot about the dowel **120** (in the clockwise direction of arrow H) towards the stationary jaw **108** and the movable handle **132** to pivot about the first pivot **126** (in the direction of arrow A) towards the stationary handle **110**. The locking angle α gradually widens, and the movable jaw **114** rotates about the dowel **120**. Thus, the jaws **108,114** move towards a closed position like that shown in FIG. **7** and begin to clamp an object. Tension in the traction spring **170** gradually increases. The tension in spring **170** pulls the teeth **204** into engagement with the teeth **214** of catch **161**. When the teeth **204** and **214** come into engagement with one another, as shown in FIGS. **7** and **8**, the operator hears the first of several audible clicks. This click is an indication that teeth **204, 214** are engaged and that an object can be firmly clamped between the jaws **108, 114**. As the operator firmly clamps the object between the jaws **108, 114**, the locking angle α slightly increases to a degree wherein the actuation rod **129** is locked. Thus, rotation of the actuation around the first pivot is permitted in a first direction (clockwise) but prevented in the opposite direction (anti-clockwise). Stresses due to clamping plus tension in the traction spring **170** tend to reduce the locking angle α . However, because of the engagement of teeth **204, 204**, the reduction of the locking angle α is prevented. Thus, the locking pliers **103** have reached the first of several stable locking or clamping positions. In this position, the trigger protrudes further beneath the movable handle **132**. The snap-fastening can be felt by the operator's index finger.

(34) If the object to be clamped is smaller, the operator can continue to move handles **110** and **132** closer to one another. Due to the fact that teeth **204** and **214** engage one another in a ratchet type fashion, additional stable locking or clamping positions can be achieved. The number of positions could theoretically be infinite. However, practical limitations such as machining tolerances and the strength of materials used in manufacturing make it preferable to have between 2 and 10 clamping positions. Even more preferable are between 2 and 7 clamping positions. See FIGS. **9** and **10**.

(35) As each incremental locking or clamping position is achieved, the moving jaw **114** moves closer to the stationary jaw **108**. In each locking or clamping position, the double torsion spring **176** urges the catch **160** to pivot in a clockwise direction E about the axle **162**. Teeth **204** and **214** engage each another with a force which is substantially perpendicular to the imaginary line IL between the first **126** and third **148** pivots. If the operator pulls the two handles **110,132** even closer together, the movable handle's abutment point **184** will move into abutment with the actuation rod's abutment point **182** before the second pivot **130** passes the imaginary line IL between the first **126** and the third **148** pivots. This guarantees that the second pivot **130** never passes the imaginary line

IL.

(36) In each of the locking or clamping positions, when the operator releases the movable handle **132**, the locking pliers **103** adopts the stable clamped position under the bias of the traction spring **170**. The lock mechanism **117** exerts a locking force LF in a clockwise direction about the dowel **120** which is counteracted by a clamping force CF between the jaws **108,114**.

(37) To release the object clamped between the jaws **108,114**, the operator takes hold of the locking pliers **103**, pulls the handles **110,132** slightly closer together until the movable handle's abutment point **184** moves into abutment with the actuation rod's abutment point **182**. The operator presses a finger on the trigger surface **218** of the catch **161** to disengage teeth **204, 214** and then releases the movable handle **132** to allow the jaws **108,114** to open automatically. This opening is caused by tension in the traction spring **170** which tends to pull the first **126** and third **148** pivots together, push the second pivot **130** away from the imaginary line IL and rotate the movable jaw **114** away from the stationary jaw **108** in the anti-clockwise direction of arrow G. The most wide-open position is delimited by abutment between the catch's stop face **216** and the actuation rod's stop face **206**. See FIGS. **11** and **12**.

(38) Thus, the elastic energy stored up upon clamping is released while the operator retains firm grip and control of the handles **110,132** and this helps to reduce, or even avoid, an explosive reaction to the hand.

(39) Referring in particular to FIG. **9**, when the stationary **110** and movable **132** handles are moved together, the operating lever **124** and actuation rod **128** abut at their mutual abutment points **182,184** and rotate a small distance in unison in the anti-clockwise direction of arrow A about the third pivot **148**. With the present invention, the single torsion spring **178** only lightly urges forward movement of the dowel **120** towards the notches **150a-150e** in the direction of arrow F. The dowel may be unseated from a notch and moved downwardly simply by an operator pulling the stationary **110** and movable **132** handles together and, in doing so, overcoming the bias of the single torsion spring **178**. Also, with the present invention, the dowel **120** may be unseated, and moved in an upward direction, simply by the operator squeezing the stationary **108** and movable **114** jaws together and, again, overcoming the bias of the single torsion spring **178**.

(40) Referring to FIG. **13**, each notched window **122a, 122b** in opposite sides of the intermediate connecting part **112** of the stationary assembly **104** is a generally elongate channel with one straight side opposite its notches **150a-150e**. The straight side assists sliding movement of the dowel between notches when the jaws are unclamped. The dowel **120** is the axis of the movable jaw **114** which, as mentioned above, acts as a trunion. The dowel is supported by the notches **150a-150e** which are shaped to resist lateral displacement of the dowel between the notches whether the jaws **108,114** be clamped or unclamped. For example, when the jaws are unclamped, the single torsion spring **178** exerts force F on the dowel **120** which is directed towards the notches **150a-150e** which support the dowel **120**. Even if the

(41) single torsion spring **178** were absent, which is an option, the notched windows **122a, 122b** are inclined forwardly so that notches **150a-150e** would tend to support the dowel **120**. When the jaws **108,114** are clamped, the force F' produced on the dowel **120** by a combination of the clamping force CF and the locking force LF is also directed towards the notches **150a-150e**.

(42) The components of the locking pliers **103** can be made of cut, stamped, pressed then assembled sheet metal. The locking pliers **103** are often used for bringing together metal objects and holding them with a view to welding them.

(43) The example and alternative embodiments described above may be combined in a variety of ways with each other. It should be noted that the present invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, the embodiments set forth herein are provided so that the disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art. Furthermore, the steps and number of the various steps illustrated in the figures may be adjusted from that

shown. The accompanying figures and attachments illustrate exemplary embodiments of the invention.

(44) Although the present invention has been described in terms of particular example and alternative embodiments, it is not limited to those embodiments. Alternative embodiments, examples, and modifications which would still be encompassed by the invention may be made by those skilled in the art, particularly in light of the foregoing teachings.

(45) Those skilled in the art will appreciate that various adaptations and modifications of the example and alternative embodiments described above can be configured without departing from the scope and spirit of the invention. Therefore, it is to be understood that, within the scope of the appended claims, the invention may be practiced other than as specifically described herein.

Claims

1. A locking pliers, comprising: a stationary assembly having a rear end forming a stationary handle and a front end forming a stationary jaw; a movable assembly having a movable jaw that is incrementally movable relative to the stationary jaw and pivotally coupled to the stationary assembly by a dowel that serves as a jaw axis which enables clamping of an object between the movable jaw and stationary jaw, the movable assembly having a latch lock mechanism including: an operating lever having a front end that is pivotally coupled to the movable jaw at a first pivot, and a rear end that forms a movable handle, an actuation rod pivotally coupled to the operating lever and the stationary handle at a second pivot, the actuation rod having a front end defining an arc, a first series of teeth formed on the arc terminating with a first stop face, the front end also defines a lower arc positioned adjacent the first stop face, and a catch arranged on and directly coupled to the operating lever, the catch including an outer side and an inner side, the outer side defining a trigger surface configured for user-engagement to selectively allow pivoting of the catch, the inner side having a second series of teeth formed on a distal end of the inner side, a second stop face formed on a proximal end of the inner side, and a recess positioned between the second series of teeth and the second stop face, the second series of teeth being configured to engage the first series of teeth to place the latch lock mechanism between one of a plurality of incremental locked positions, and allow rotation of the actuation rod which incrementally moves the movable jaw towards the stationary jaw in a manner that holds the movable jaw and the stationary jaw in a fixed, clamped position; and wherein in an open, unclamped position, the first series of teeth and the second series of teeth are completely disengaged and the recess provides clearance to allow the first series of teeth to avoid contact with the catch, while simultaneously permitting the first stop face to be in the direct engagement with the second stop face to thereby prevents the stationary handle and the movable handle from moving further apart.
2. The locking pliers of claim 1, wherein the first series of teeth and the second series of teeth are configured to engage each other to permit multiple clamping positions in which the movable jaw and the stationary jaw are held in a fixed, clamped arrangement.
3. The locking pliers of claim 1, wherein the lower arc has a radius that is less than a radius of the upper arc.
4. The locking pliers of claim 1, wherein the engagement of the first series of teeth and the second series of teeth allows rotation of the actuation rod in a first direction that establishes the clamped position between the movable jaw and the stationary jaw.
5. The locking pliers of claim 4, wherein the second stop face is configured to engage first stop face and prevent further rotation of the actuation rod in the first direction.
6. The locking pliers of claim 4, wherein rotation of the actuation rod in a second direction opposite to the first direction moves the stationary jaw and the movable jaw from a clamped position to said open, unclamped position.
7. The locking pliers of claim 6, wherein in the open, unclamped position, the first stop face is in

direct engagement with the second stop face.

8. The locking pliers of claim 1, wherein in the open position, the direct engagement of the first stop face and the second stop face delimits a most wide-open position of the stationary jaw and the movable jaw.

9. The locking pliers of claim 1, wherein the engagement of the first series of teeth and the second series of teeth produces an audible click that an object can be clamped between the stationary jaw and the movable jaw.
